

Deep Learning for Content Moderation

on the Shareish Solidarity Platform

Seyfullah Ural

October 28, 2025

Master's Thesis Progress Report

Project Overview & Objectives

Research Goals

- Design AI-based moderation for Shareish platform
- Address cold-start problem with limited data
- Ensure French language support
- Balance accuracy, fairness, and explainability

Key Challenges

- No existing Shareish moderation dataset
- Limited French content moderation research
- Need for customizable safety taxonomy

34 Papers Analyzed

Comprehensive review covering LLM-based approaches, traditional ML, datasets, and evaluation frameworks

Key Findings

- **LLM-based models** are state-of-the-art
- **Multi-dimensional evaluation** essential beyond accuracy
- **French support** critical but underexplored (Usually byproduct of LLM capability)

Key Papers: Influential Research

LLM-Based Approaches

- **Llama Guard 3** (Meta, 2024)
- **ShieldGemma** (Google, 2024)
- Watch Your Language (Kumar et al.)
- Adapting LLMs (Chinese researchers)

Traditional ML

- **Detoxify** (BERT-based, 2020)
- OpenAI Moderation API (2022)
- Perspective API (Google Jigsaw)
- CNN/LSTM/BERT comparisons

Critical Datasets

ToxiGen (274K adversarial examples), **HateCheck French** (2K test cases),
Multilingual Reddit (20K French samples)

Traditional ML vs. LLM Approaches

Traditional ML

CNN, LSTM, BERT-based classifiers

✓ Advantages

- Fast inference (50-200ms)
- Lower computational cost
- Proven reliability

× Limitations

- Limited context understanding
- Fixed categories
- No explanations
- F1 score: 0.70-0.85

LLM-Based

Llama Guard 3, ShieldGemma, Mistral

✓ Advantages

- Customizable taxonomy
- Generates explanations
- Better context awareness
- F1 score: 0.75-0.85

× Limitations

- Slower (200-500ms)
- Higher GPU requirements
- More complex deployment

Cold Start Problem & The Challenge

! The Challenge

Shareish has **no existing moderation dataset** — we need a strategy that works with limited labeled data.

Two Approaches for Training

Option A: Safety-Specialized

(*Llama Guard 3*)

- Already fine-tuned for moderation
- Ready to use out-of-the-box
- Minimal adaptation needed

+ Fast deployment
+ Lower risk

Option B: General Base Model

(*Llama 3.1-8B, Mistral 7B*)

- Pre-trained, but not for moderation
- Requires safety fine-tuning
- Full customization possible

+ More customizable
- Requires more compute

Training Strategies: Step-by-Step

Option A Strategy

1. Start with Llama Guard 3
(*pre-trained + safety fine-tuned*)
2. Fine-tune on French datasets
(*10K-50K examples*)
3. Active learning with Shareish
(*continuous improvement*)

Option B Strategy

1. Start with Llama 3.1-8B
(*general pre-trained model*)
2. Fine-tune on ToxiGen
(*274K adversarial examples*)
3. Fine-tune on French datasets
(*10K-50K examples*)
4. Active learning with Shareish
(*continuous improvement*)

Question for Supervisors

Which training approach would you recommend: Option A, Option B, or both for comparative analysis?

Evaluation Framework: Beyond Accuracy

Technical Performance

- F1 Score (Precision + Recall)
- HateCheck pass rate (>70%)
- AUC-ROC

Operational Metrics

- Inference latency
- Cost per 1000 requests
- GPU requirements

Fairness & Legitimacy

- False Positive Rate disparity
- Explainability

Two-Tier Hybrid Approach

STAGE 1: Fast Pre-filter Detoxify (BERT)

- Inference: $\sim 50\text{ms}$
- Filter obvious violations
- Pass borderline cases

STAGE 2: Detailed Analysis Llama Guard 3 (LLM)

- Inference: $\sim 200\text{-}300\text{ms}$
- Customizable taxonomy
- Generates explanations

Why This Approach?

Balances **speed** (fast filter for most content), **accuracy** (LLM for complex cases), and **cost** (only $\sim 10\text{-}20\%$ reach Stage 2).

Alternative: Single-step LLM for maximum simplicity (under evaluation)

Timeline Overview (20 Weeks)

Weeks 1-4 Model Selection

- Compare candidates
- Test architectures
- Baseline testing

Weeks 5-12 Implementation

- Build system
- Fine-tune models
- Comprehensive evaluation

Weeks 13-20 Thesis Writing

- Draft all chapters
- Document results
- Finalize manuscript

Target completion: April 2026
(3-month buffer before June deadline)

Current Status & Immediate Next Steps

What's Been Accomplished

✓ Completed

- Literature review (34 papers)
- Candidate models identified
- Datasets accessible (ToxiGen, HateCheck French)
- Evaluation framework designed

→ This Month (Weeks 1-4)

- Comparative model testing
- Architecture decision
- Learn Alan GPU cluster tools
- Begin baseline implementation

Ready to Begin

All prerequisites complete — ready to start the comparative testing phase immediately

Questions for Discussion

- Shareish Requirements: Daily message volume? Acceptable latency?
 - Data Access: Historical moderation examples available?
 - Content Taxonomy: Final definition of “toxic content” for Shareish?
 - Scope: Text only, or include images later?, implement every little improvement?
-

Thank You!

Ready to begin comparative testing phase